

**DATE : 11.07.2025**

**DT/NT : DT**

**LESSON : DEEP LEARNING**

**SUBJECT: RNN- LONG SHORT-TERM  
MEMORY(LSTM) - GRU**

**BATCH : 287**

**DATA  
SCIENCE**



**TECHPRO**  
EDUCATION



techproeducation.com

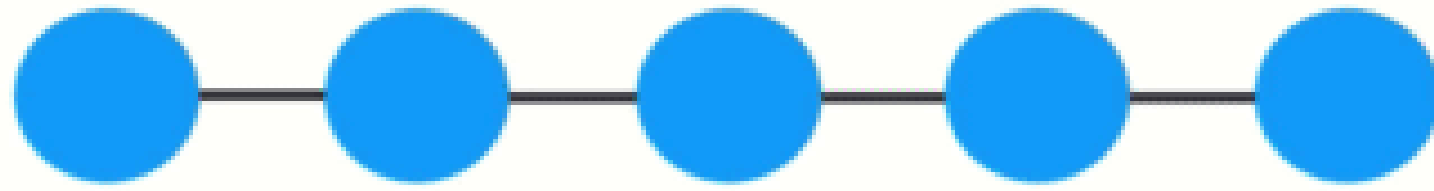


+1 (585) 304 29 59



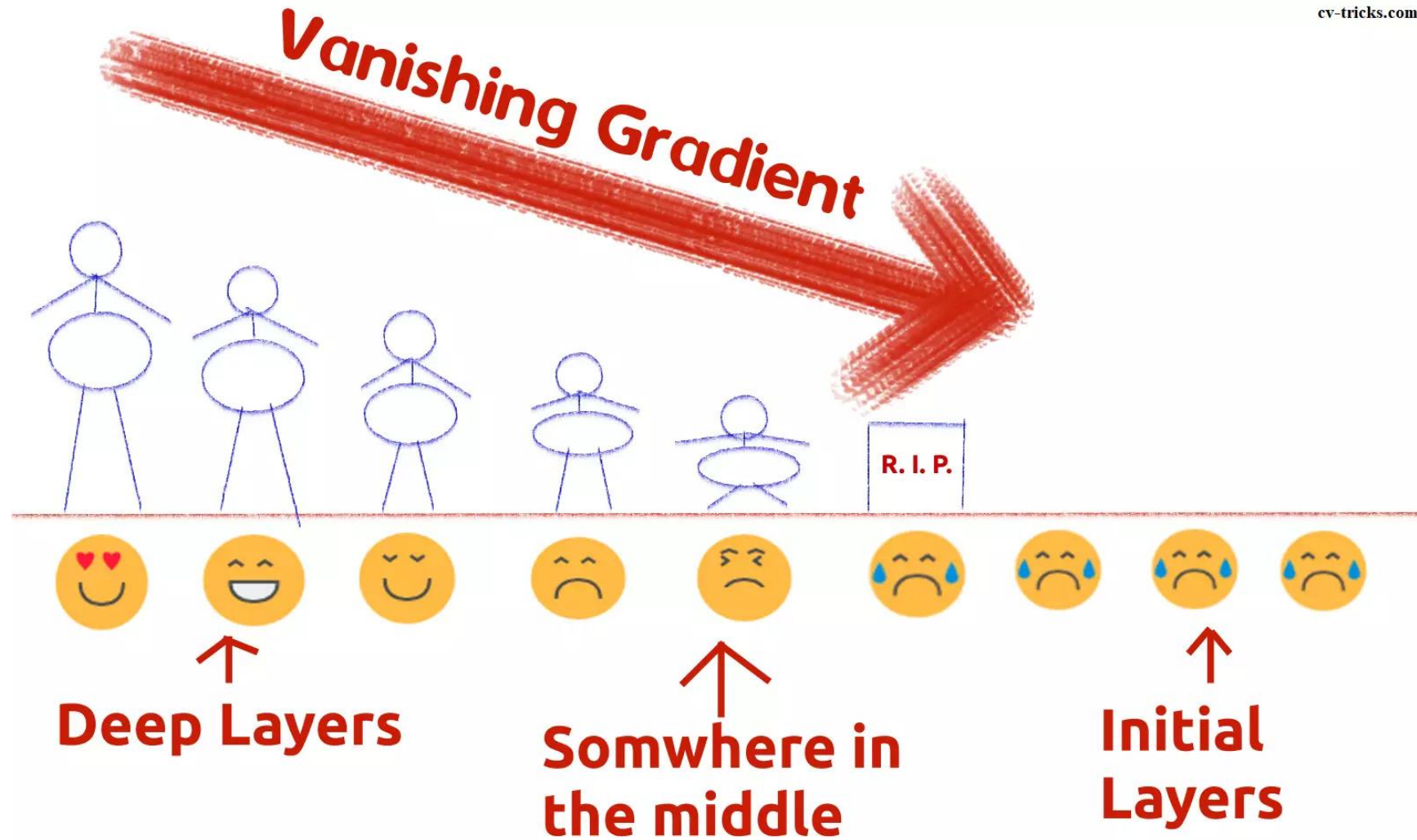


# **VANISHING /EXPLODING GRADIENT PROBLEM**



# RNN VANISHING /EXPLODING GRADIENT

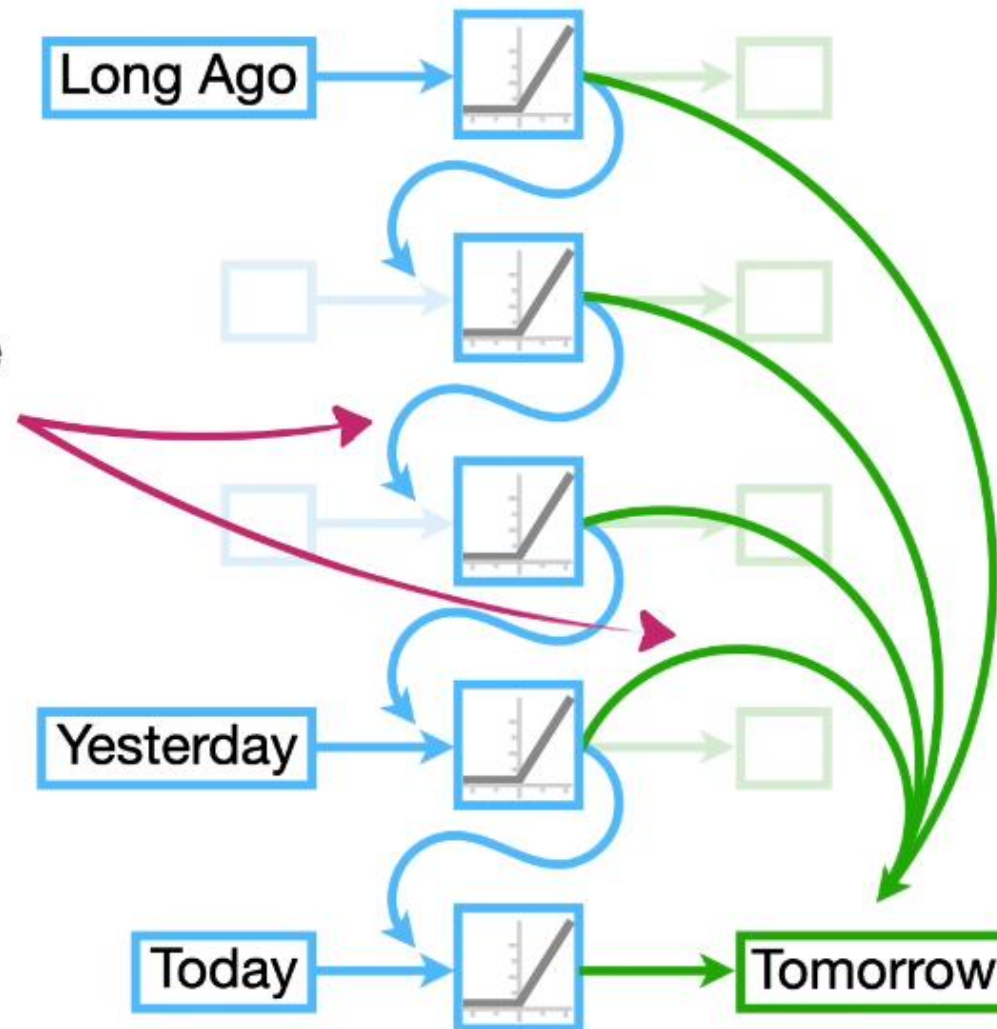
cv-tricks.com



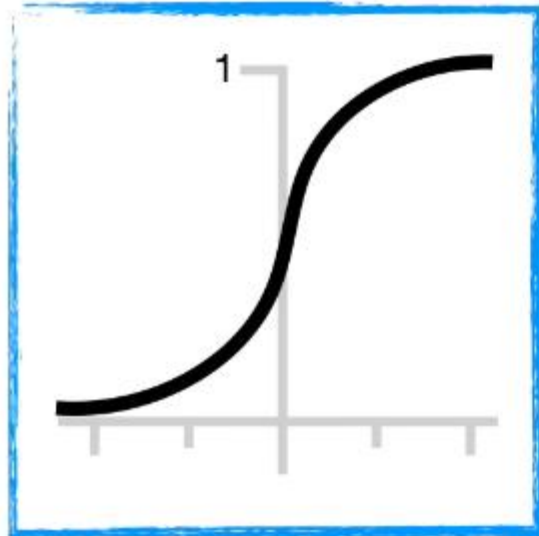
The background features abstract, flowing curved lines in shades of green and grey. A thick grey line curves from the top left towards the center. Another thick green line curves from the top left towards the bottom left. A large, thick green line curves from the top right towards the bottom right, forming a large, open 'C' shape on the right side of the image.

# **LSTM (LONG SHORT- TERM MEMORY)**

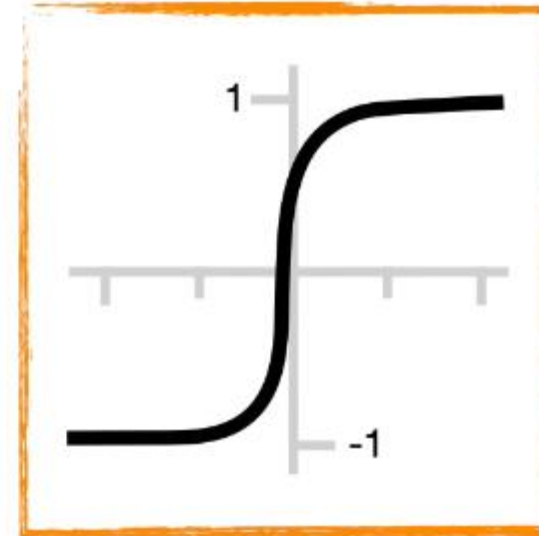
...**Long Short-Term Memory** uses two separate paths to make predictions about tomorrow.



So, now that we know that the **Sigmoid Activation Function** turns any input into a number between **0** and **1**...



...and the **Tanh Activation Function** turns any input into a number between **-1** and **1**...



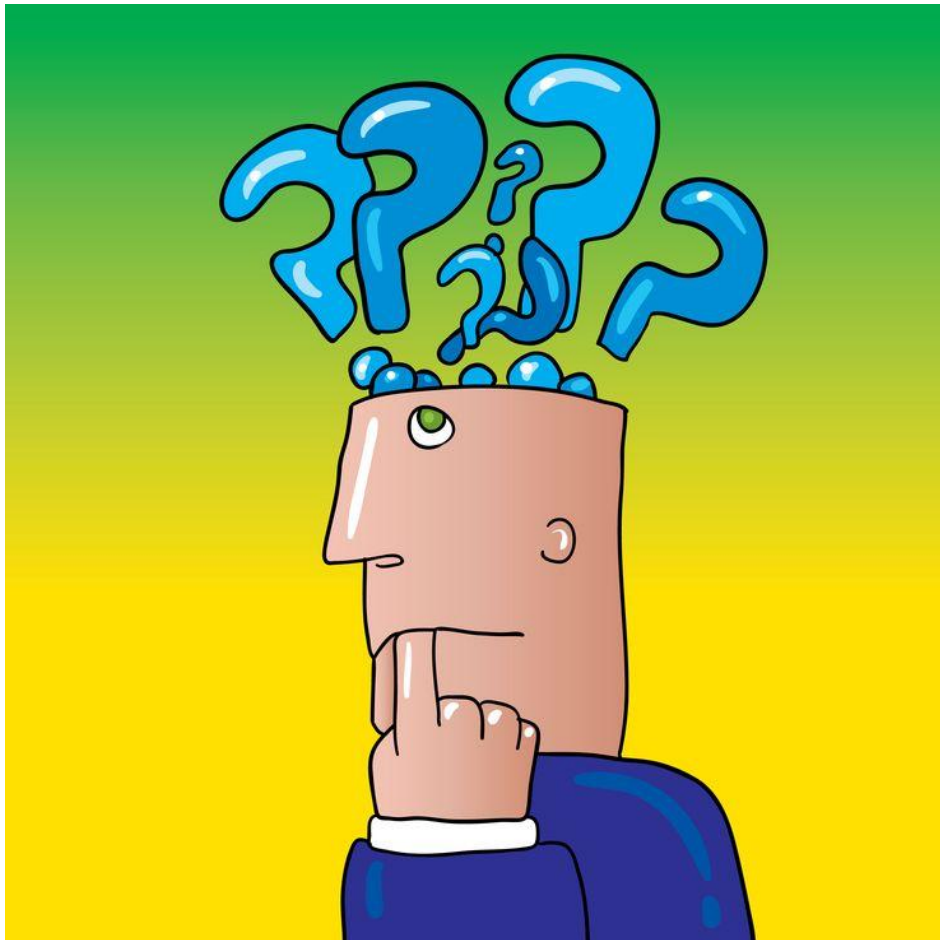


# What is LSTM?





"I will lead the way to success.."  
"She wore a necklace made of lead."



## Customers Review 2,491

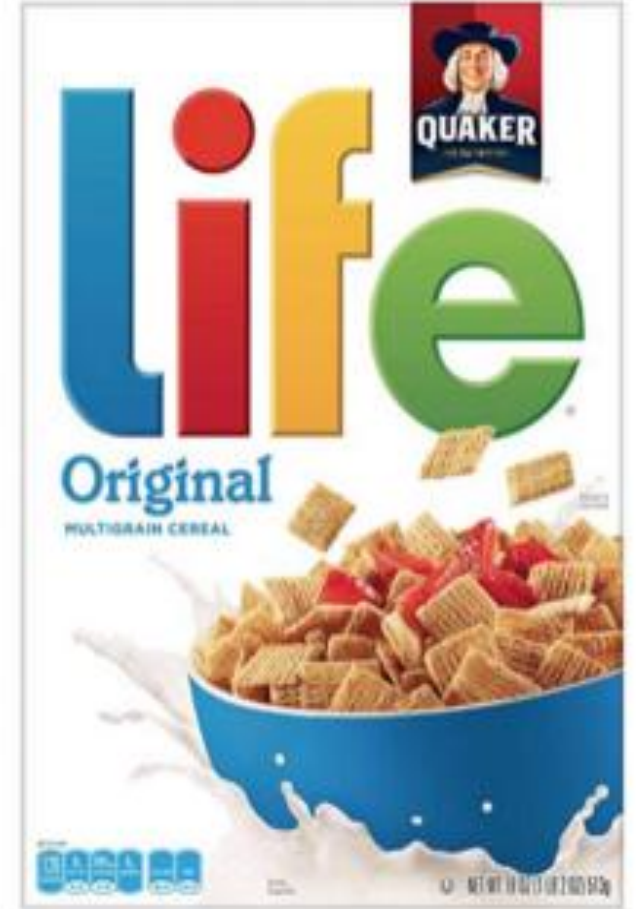


Thanos

September 2018

Verified Purchase

**Amazing! This box of cereal gave me a perfectly balanced breakfast, as all things should be. I only ate half of it but will definitely be buying again!**



**A Box of Cereal**

**\$3.99**

## Customers Review 2,491

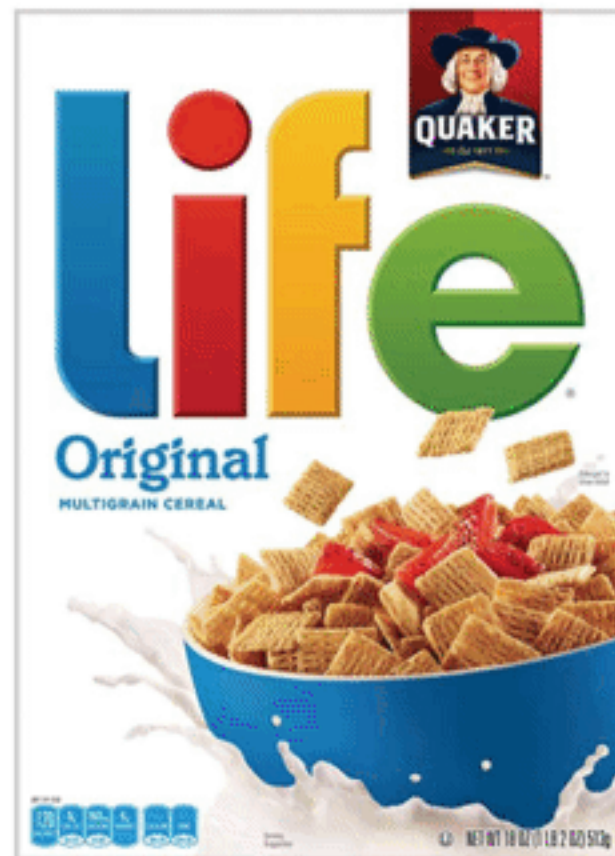


Thanos

September 2018

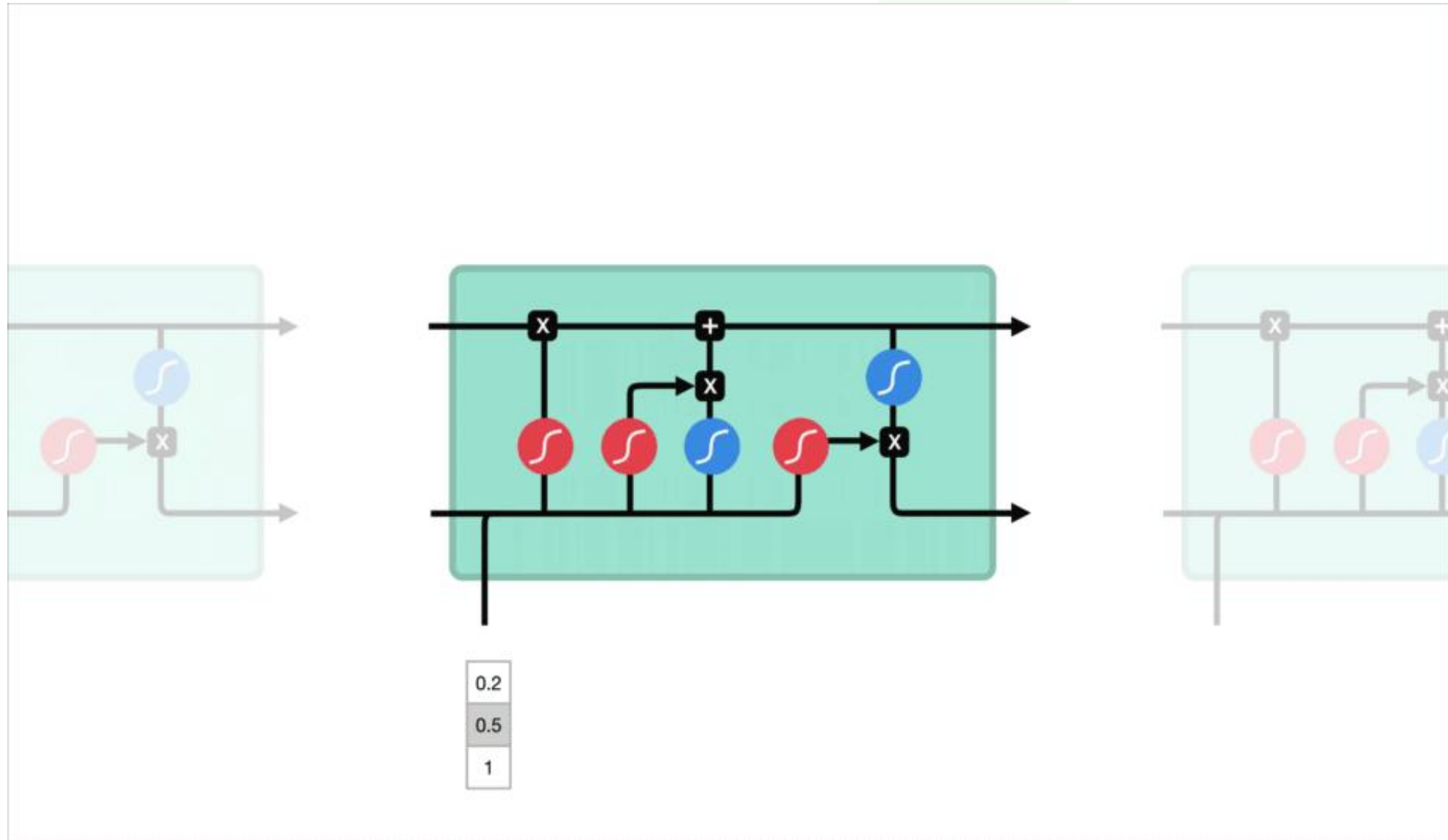
Verified Purchase

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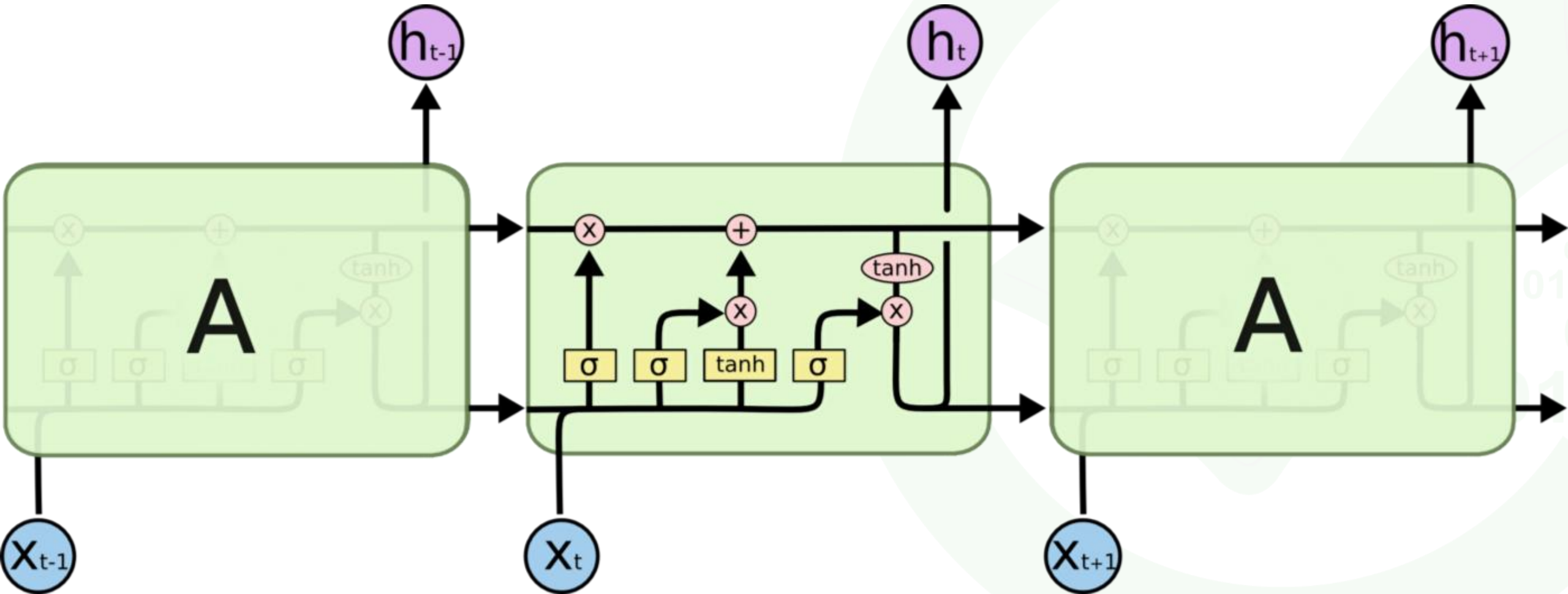


**A Box of Cereal**  
**\$3.99**

# LSTM

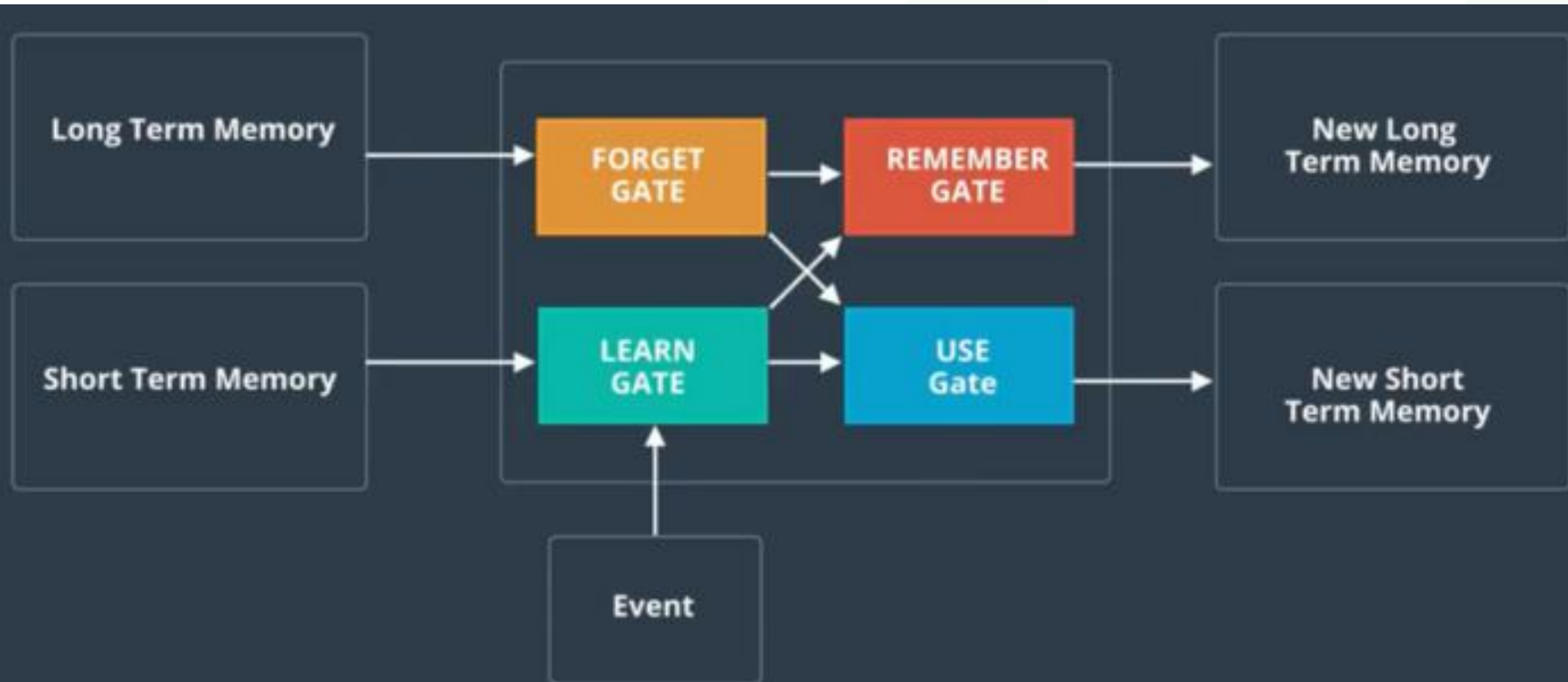


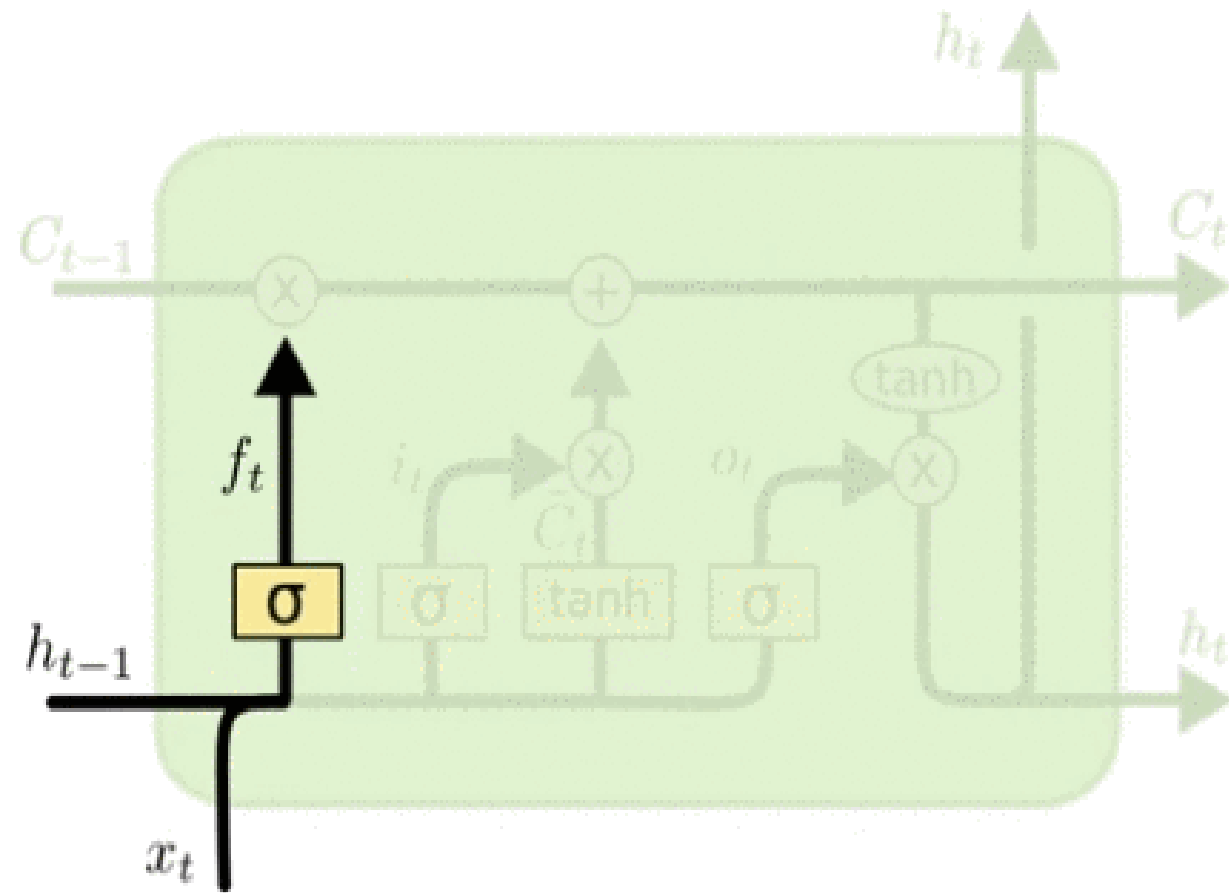
# LSTM





# LSTM

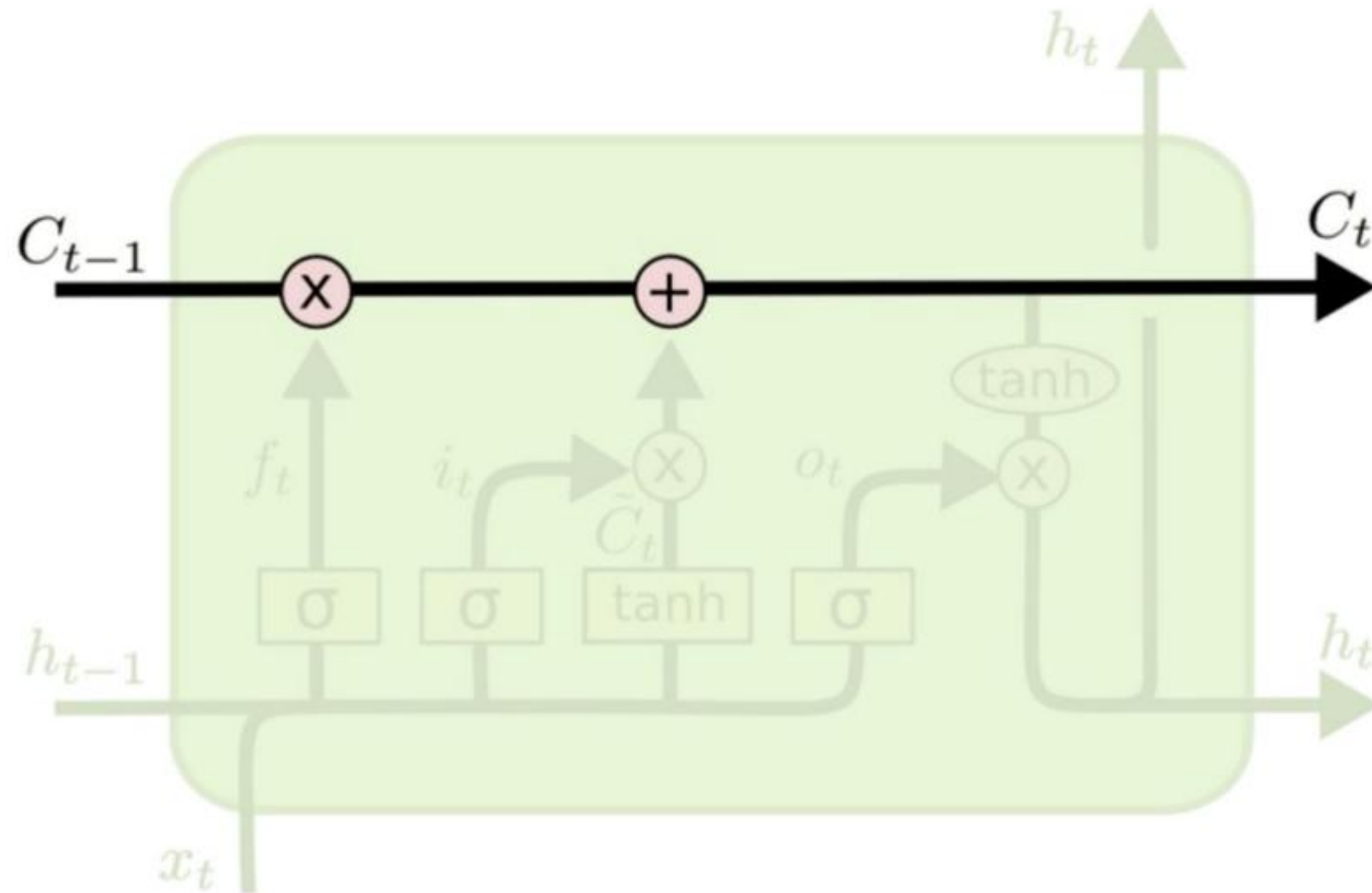




$$(1) f_t = \sigma (W_f \cdot [h_{t-1}, x_t] + b_f)$$



# LSTM



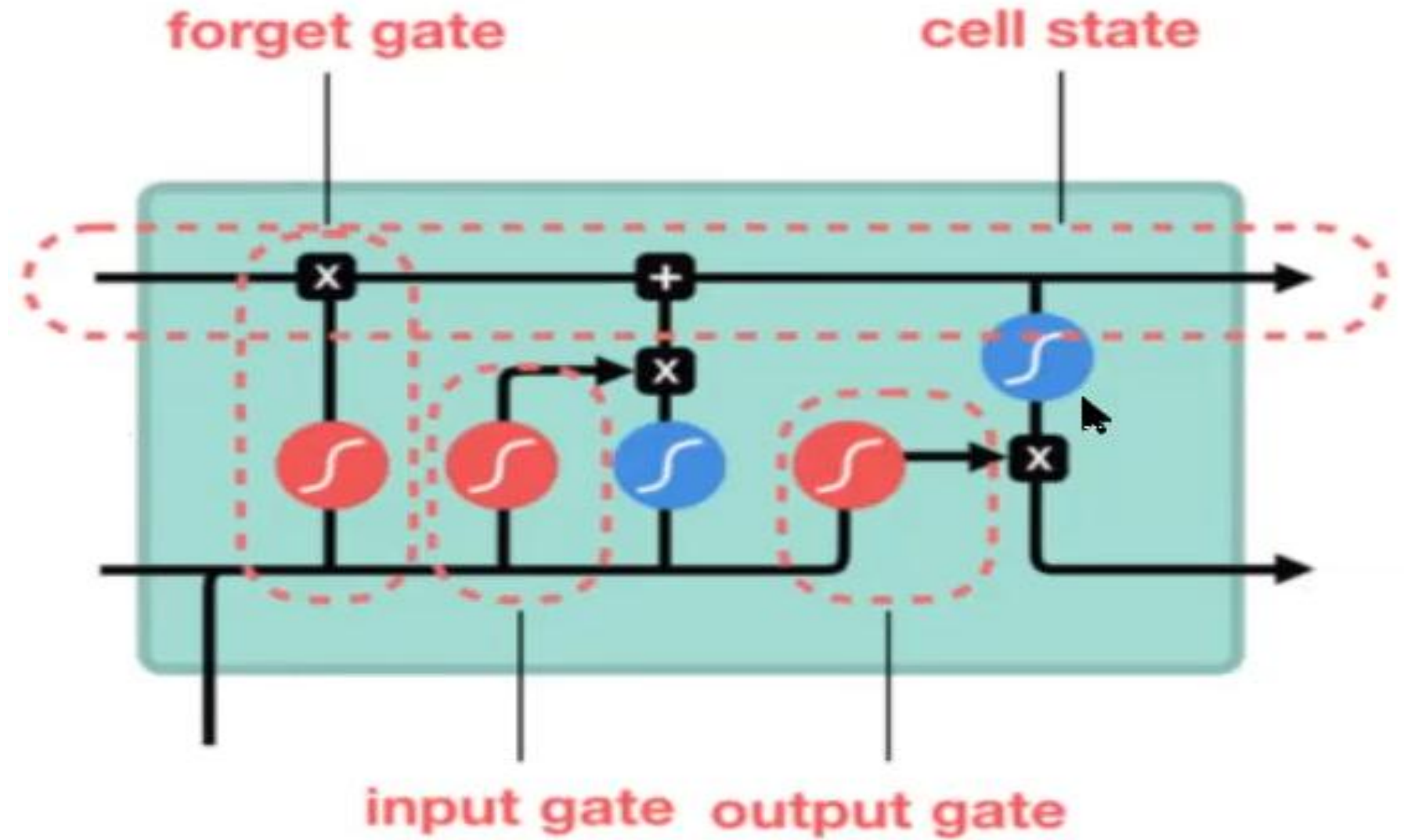
# LSTM



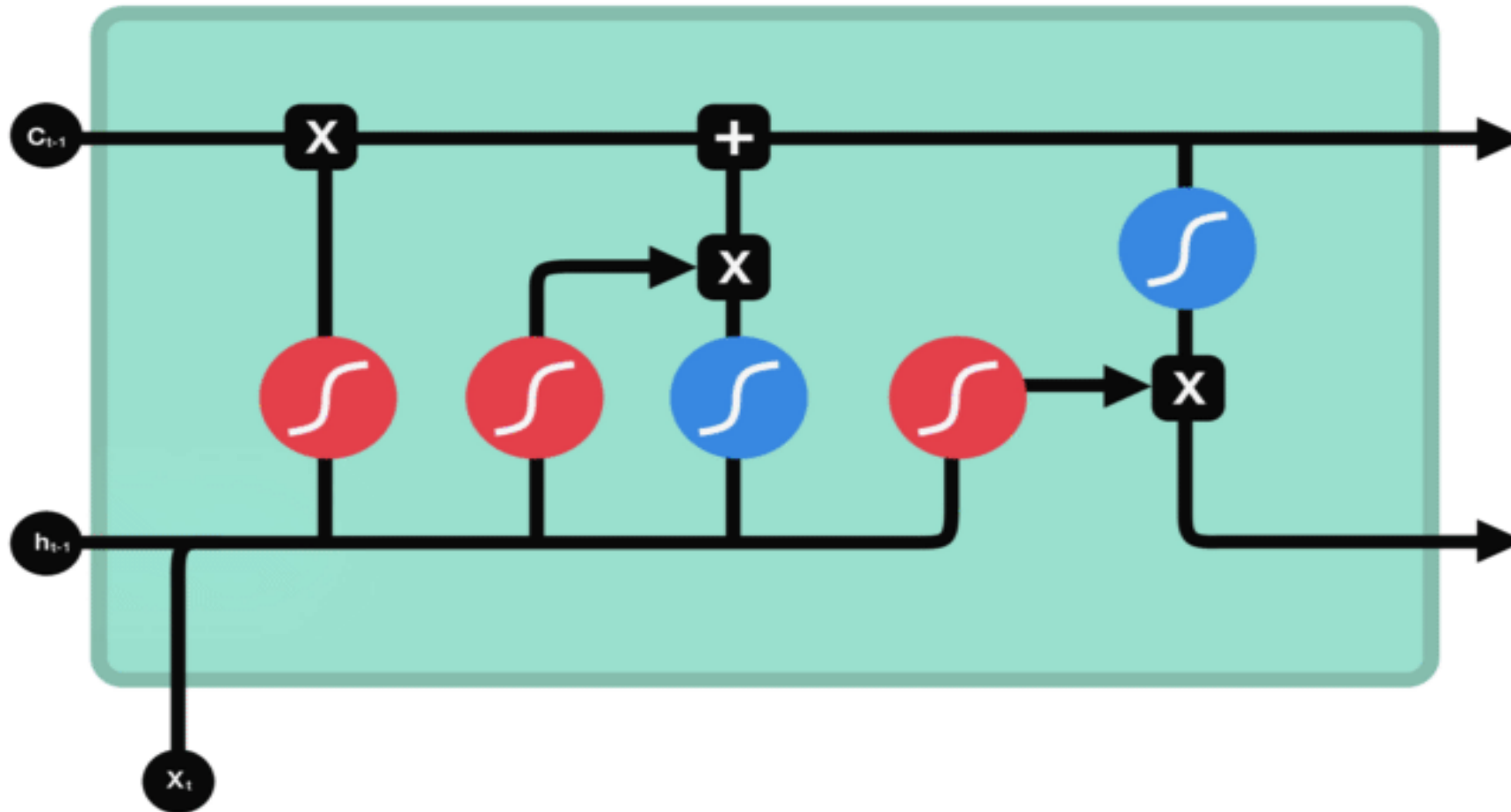
sigmoid



tanh

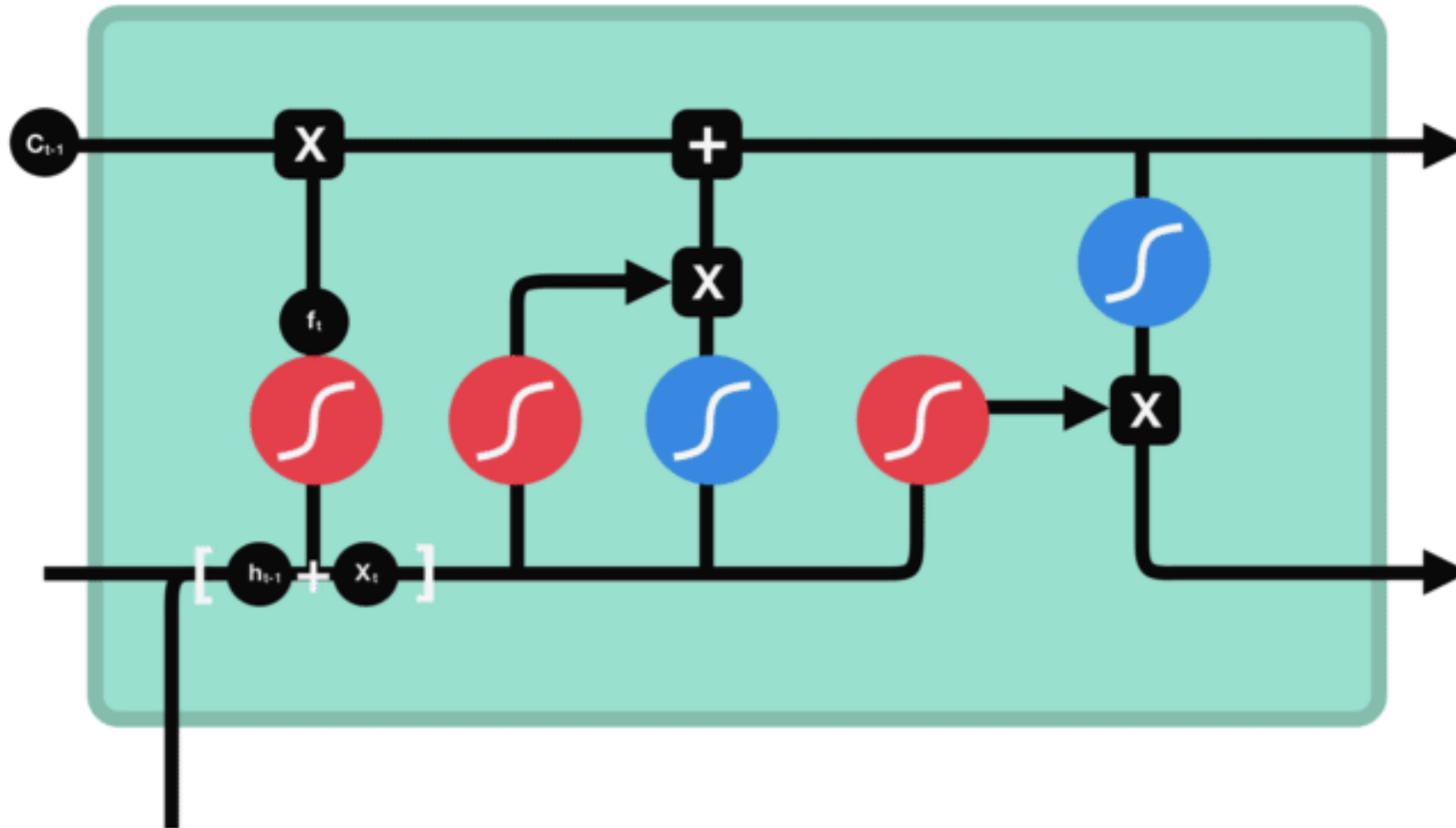


## 1-) Forget Gate



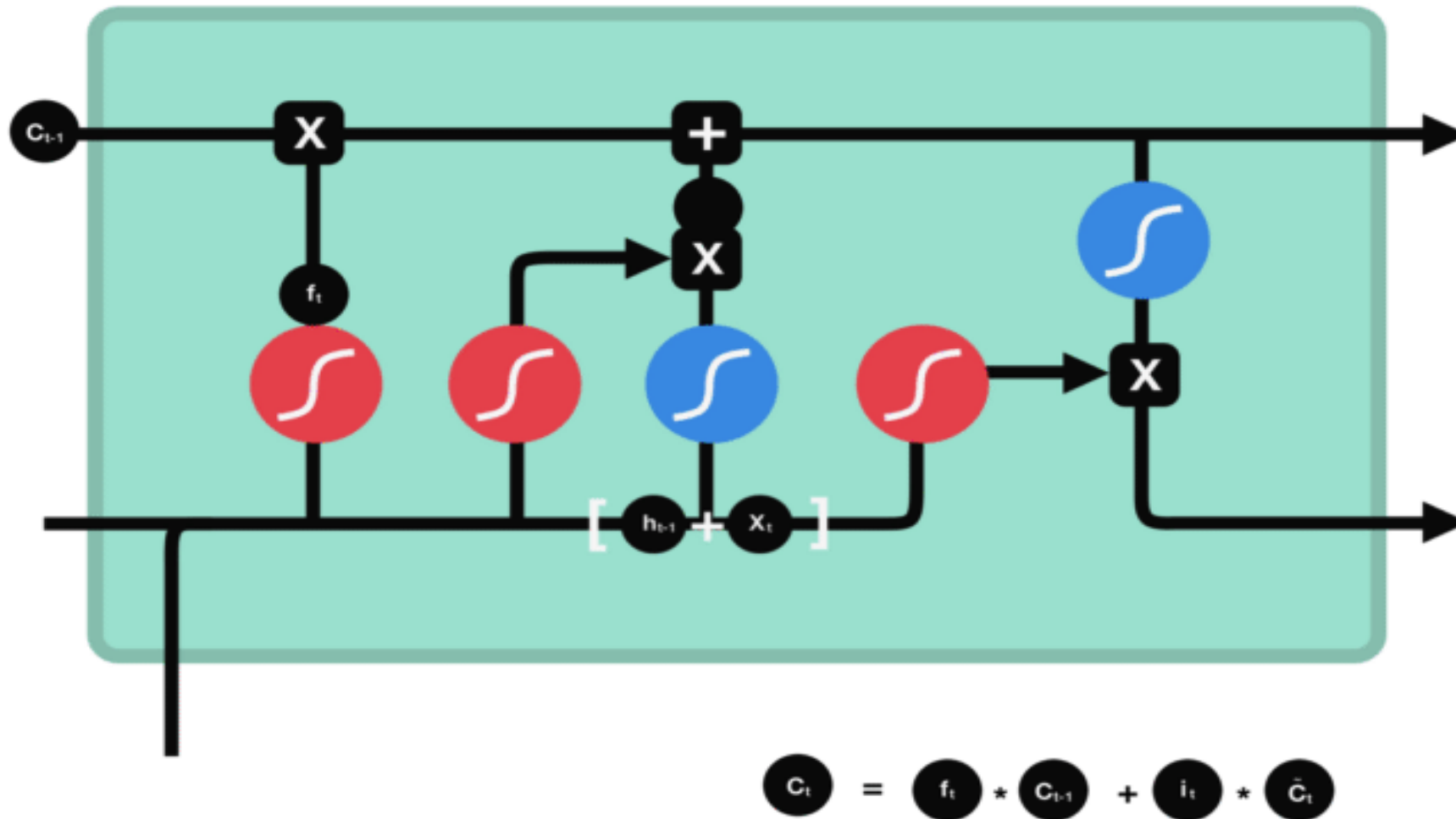
$c_{t-1}$  previous cell state  
 $f_t$  forget gate output

## 2-) Input Gate



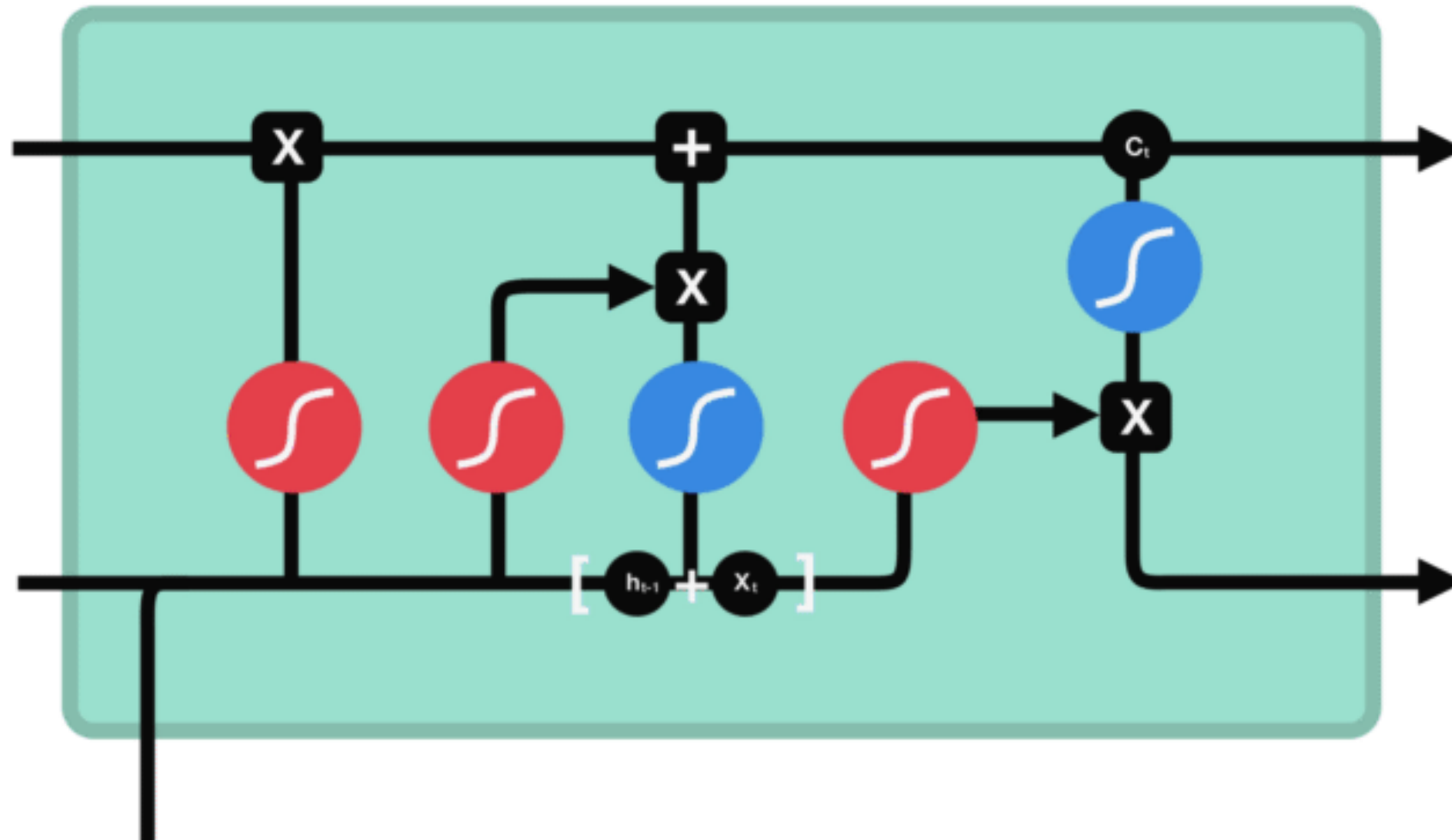
- $C_{t-1}$  previous cell state
- $f_t$  forget gate output
- $i_t$  input gate output
- $\tilde{C}_t$  candidate

## 3-) Cell State



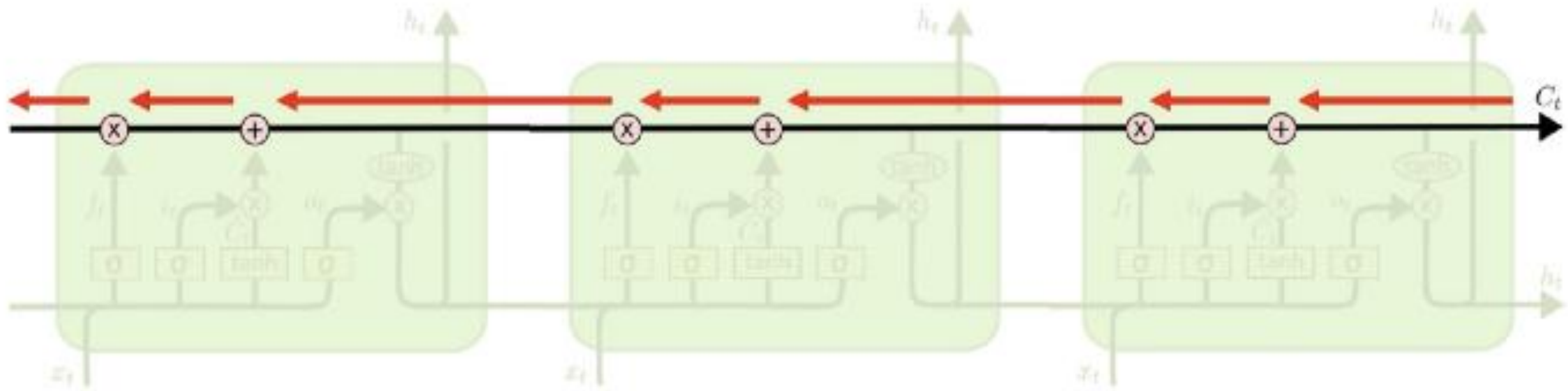
- $C_{t-1}$  previous cell state
- $f_t$  forget gate output
- $i_t$  input gate output
- $\tilde{C}_t$  candidate
- $C_t$  new cell state

## 4-) Output Gate



- $c_{t-1}$  previous cell state
- $f_t$  forget gate output
- $i_t$  input gate output
- $\tilde{c}_t$  candidate
- $c_t$  new cell state
- $o_t$  output gate output
- $h_t$  hidden state

# LSTM - BACKPROPAGATION

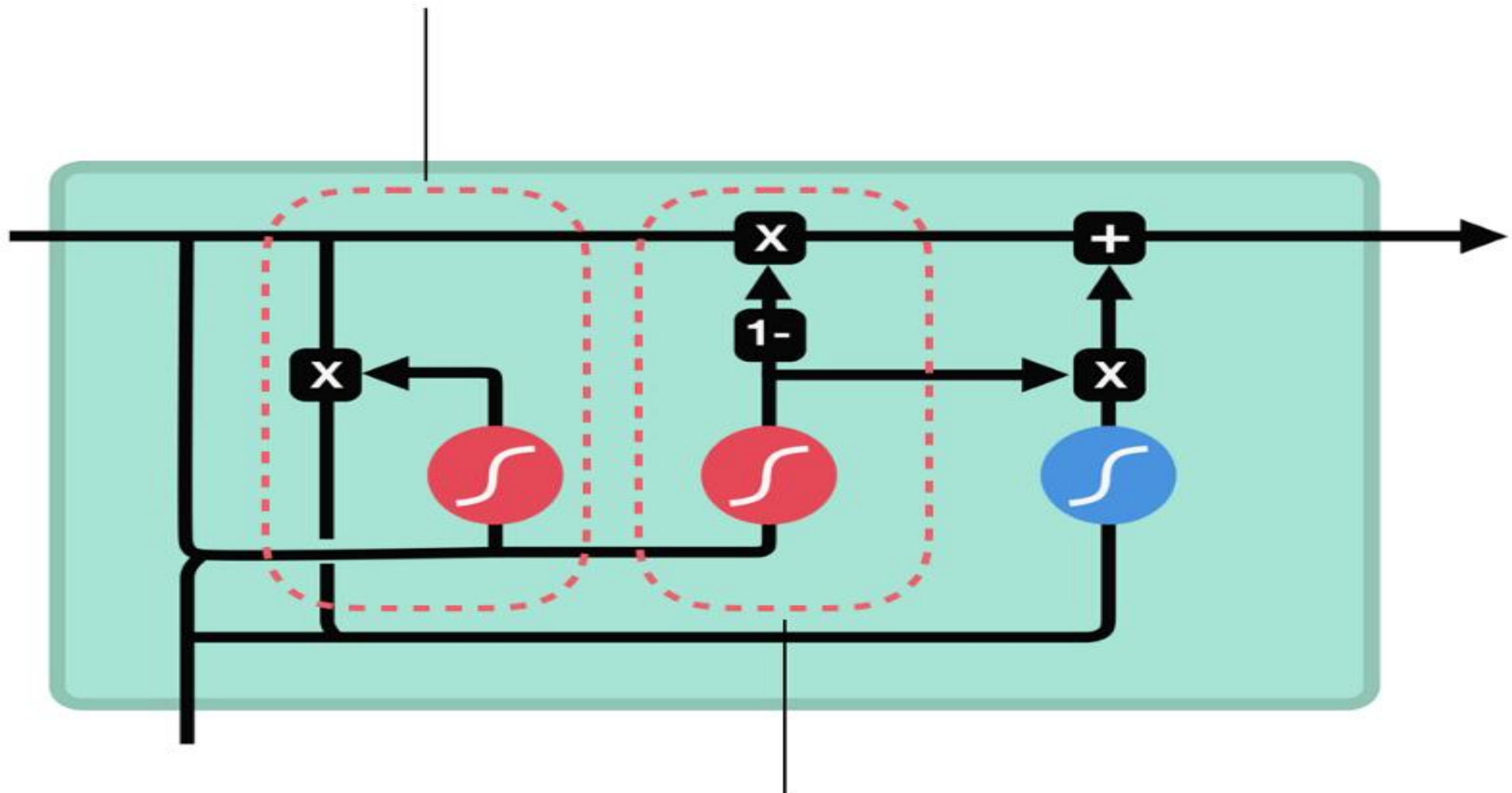






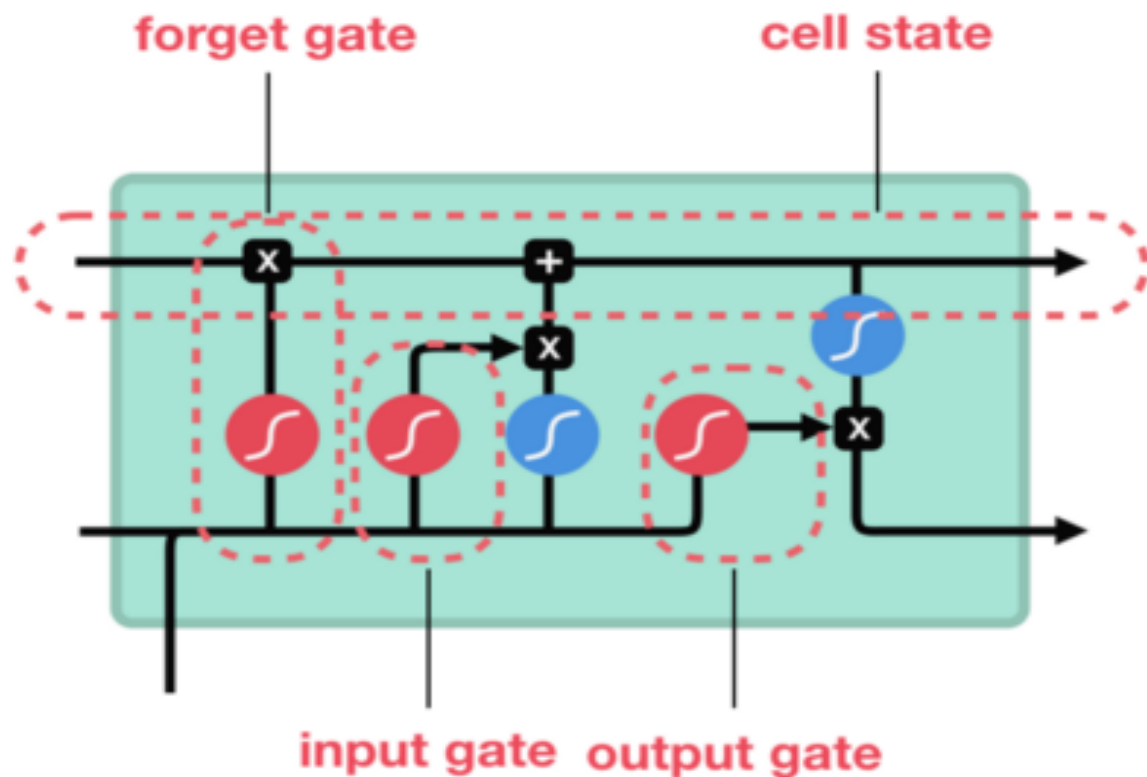
**GRU (GATED  
RECCURENT UNITS)**

reset gate

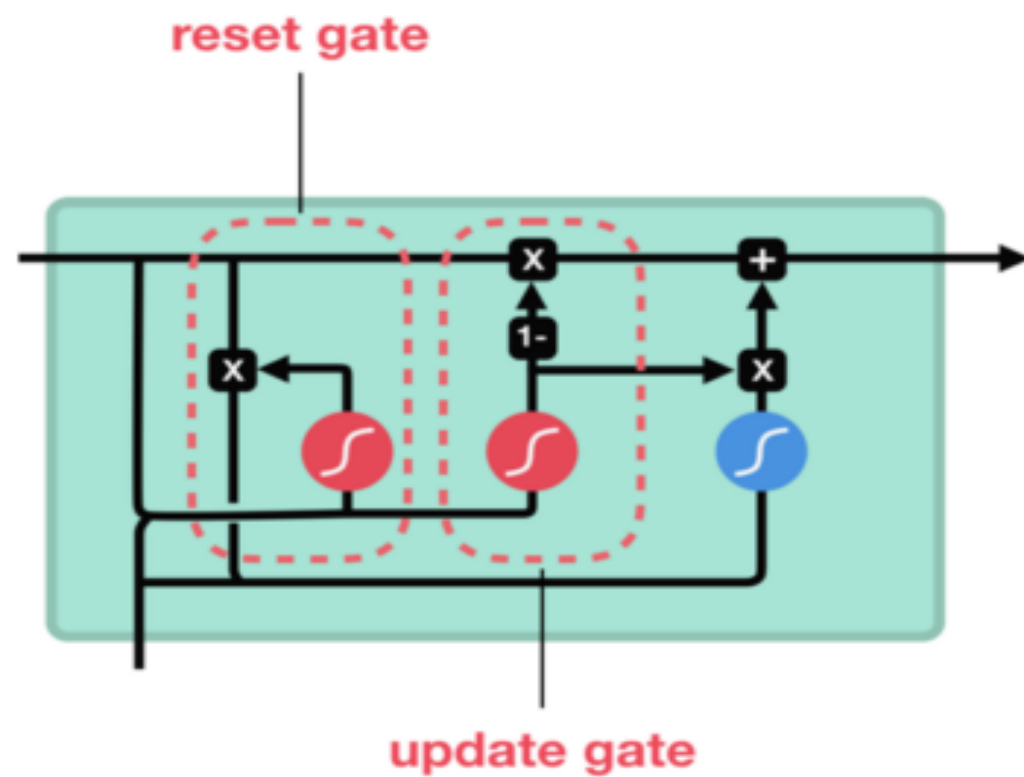


update gate

## LSTM



## GRU



sigmoid



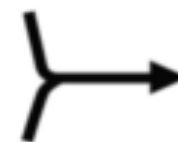
tanh



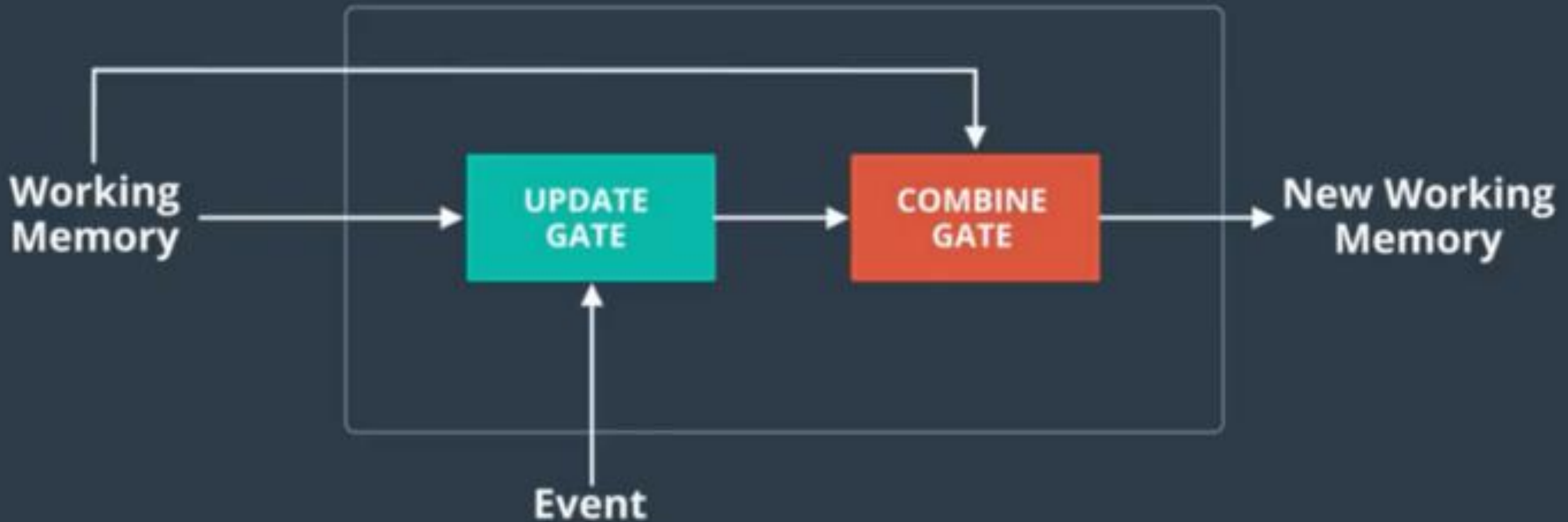
pointwise  
multiplication



pointwise  
addition



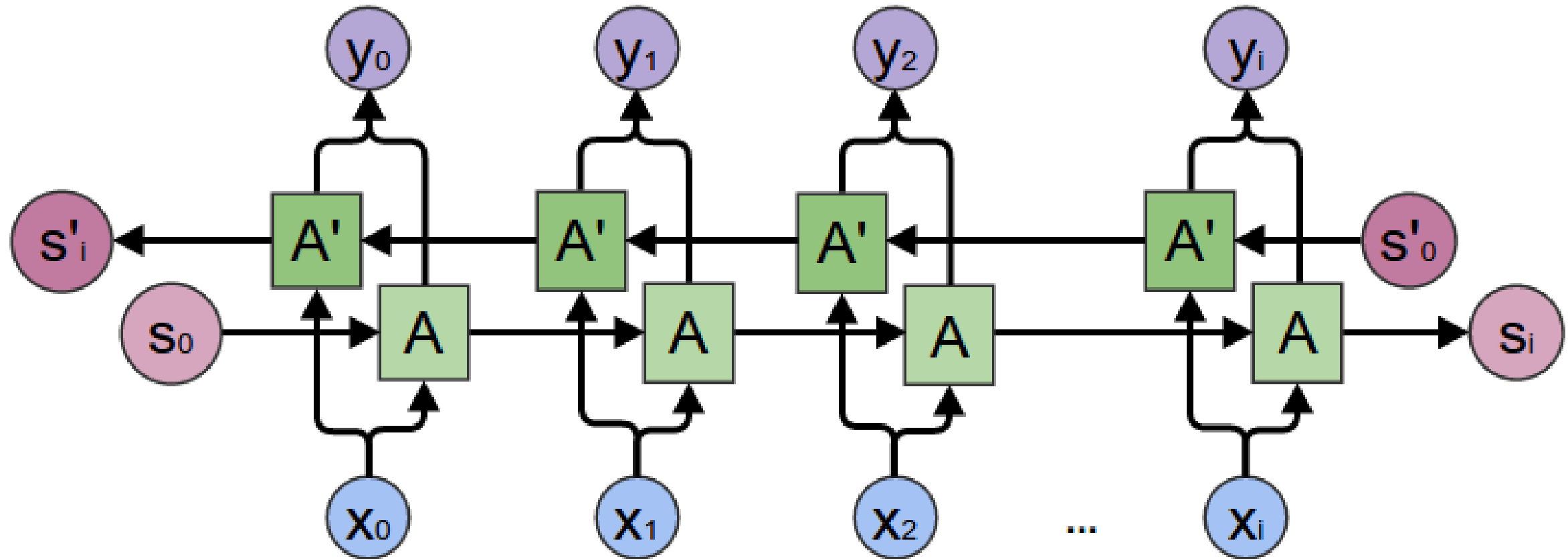
vector  
concatenation





**BIDIRECTIONAL  
RNN/LSTM/GRU**

# BIDIRECTIONAL RNN/LSTM/GRU



Heart is not enlarged

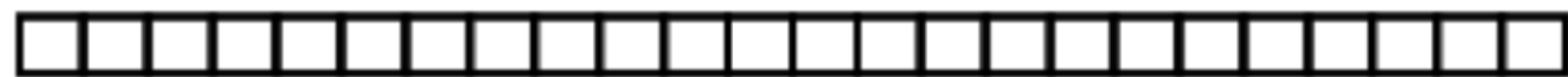
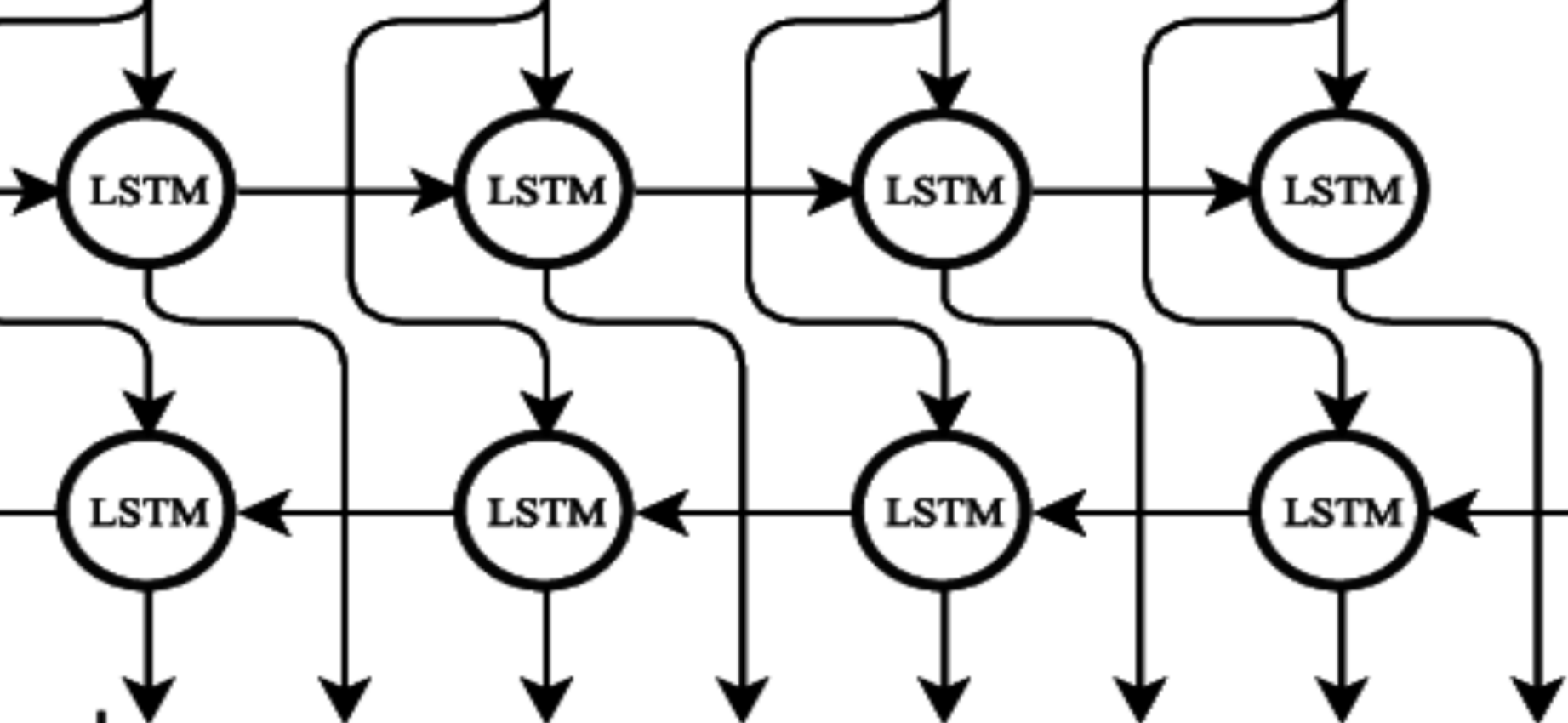
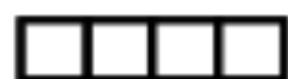
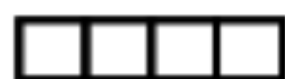
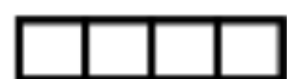
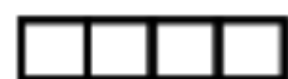
Embedding Layer

Forward  
LSTM

0

Backward  
LSTM

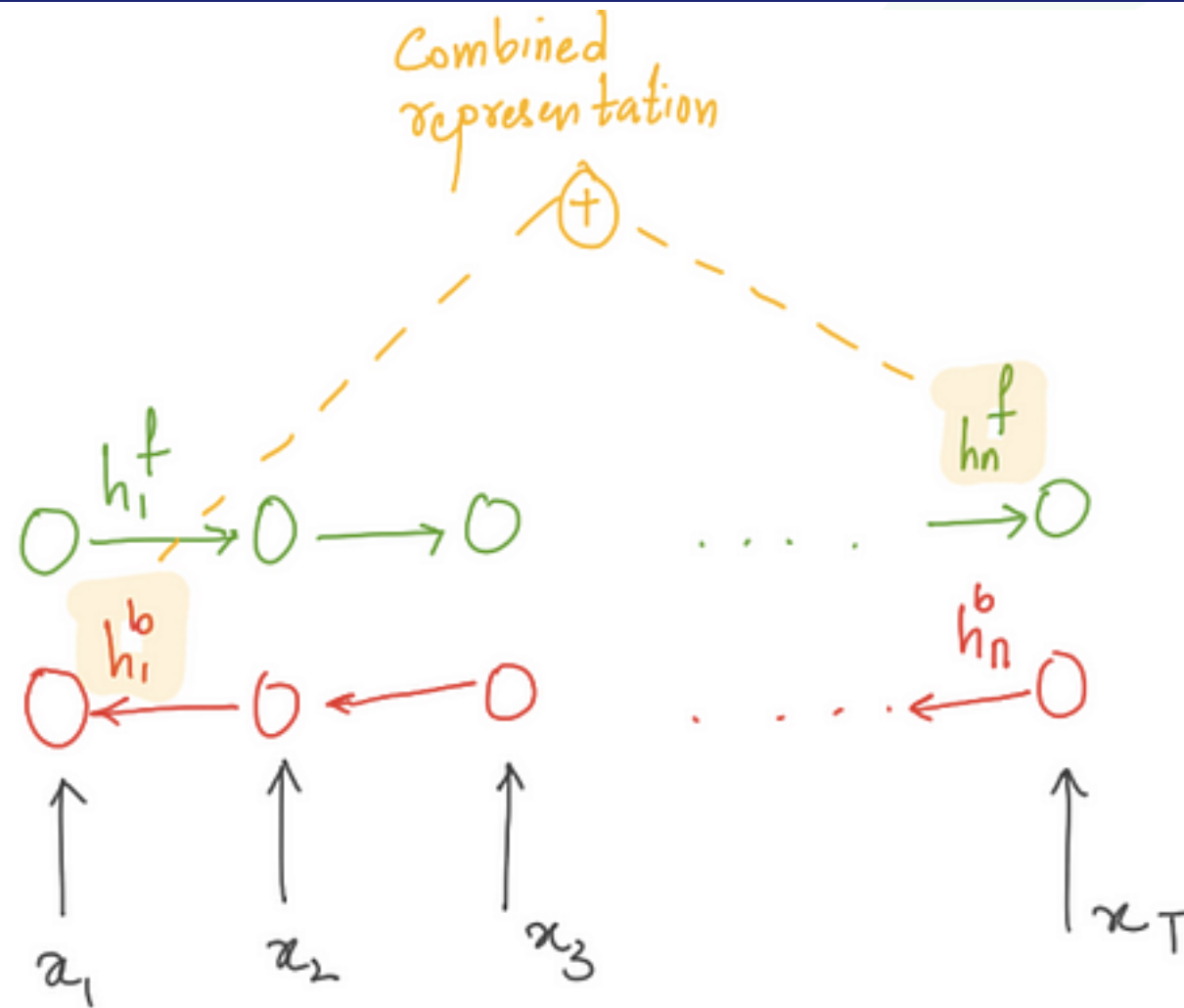
Concatenate &  
Flatten



8

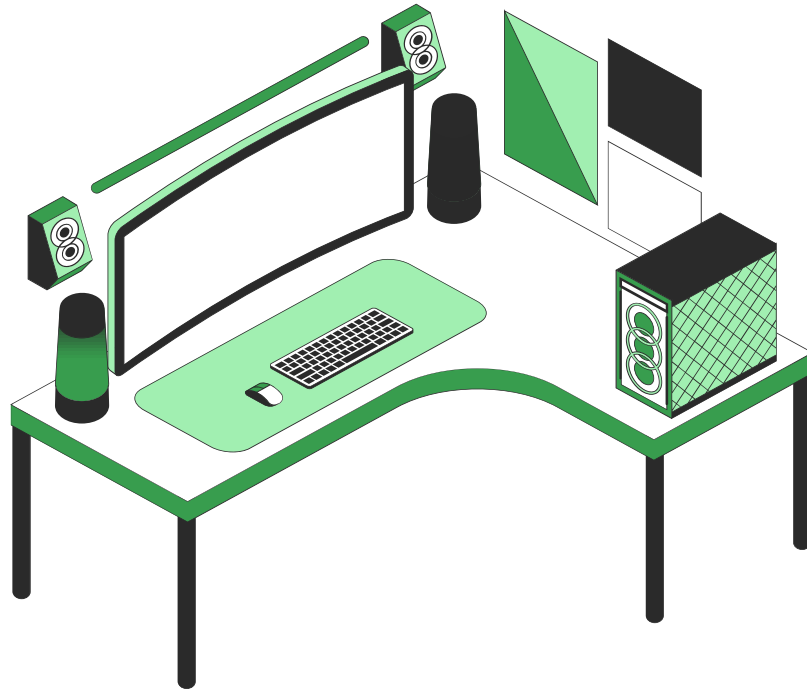


# BIDIRECTIONAL RNN/LSTM/GRU



RNN forward  
RNN backward

# THE END



## Do you have any questions?

Send it to us! We hope you learned something new.

# 10:00

Start

Stop

Reset

mins:

secs:

type: 

None

▼